DEEP LEARNING CIE SUBMISSION

PBL TOPIC: ATTENTION-BASED TEXT CLASSIFICATION

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Project Topic:	Attention-Based Text Classification
Paper Reviewed:	*Effective Approaches to Attention-based Neural Machine Translation*
Authors & Year:	Minh-Thang Luong, Hieu Pham, Christopher D. Manning (2015)
Publication Venue:	EMNLP 2015 | arXiv:1508.04025

1. Problem Addressed

Compares global vs local attention mechanisms and multiplicative vs additive score functions to improve computational efficiency.

2. Motivation & Need

Global attention requires inspecting all encoder hidden states for every target token, which is expensive for very long sequences.

3. Methodology & Model Architecture

Proposed Global Attention (attending to all source tokens) and Local Attention (attending to a small window centered at predicted alignment position p_t).

4. Dataset & Experimental Setup

WMT 2014 and WMT 2015 English-to-German and English-to-French translation datasets.

5. Attention Mechanism Formulation

Multiplicative Dot-Product Score: score(h_t, h_s) = h_t^T * W_a * h_s. Demonstrates fast GPU matrix multiplication.

6. Key Results Reported in Paper

Achieved a 5.0 BLEU point improvement over non-attention translation baselines on WMT English-German translation.

7. System Advantages & Strengths

Simplified score computation using dot products and reduced compute overhead via local window context alignment.

8. Limitations & Constraints

Local attention requires hyperparameter tuning for window size D and alignment position predictor.

9. Identified Research Gap

Lack of comparative study between global and local attention mechanisms in neural sequence models.

10. Relevance to Our PBL Project

Helps compare dot-product multiplicative scoring vs additive tanh scoring for sequence classification.

11. Student's Critical Observations

Luong attention highlighted that dot-product attention executes faster on modern GPU tensor cores than additive tanh attention.

12. References

[1] M.-T. Luong et al., 'Effective approaches to attention-based neural machine translation,' EMNLP 2015.